

Manasvi R Kotian

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Professional Summary

Backend-oriented software engineer with 2+ years of professional experience building and operating production database services, currently developing microservice-based REST APIs with Python (FastAPI), Docker and uv. Strong grounding in testing, continuous integration, code reviews and Agile delivery, backed by a 6-month Java development certification and relational database experience.

Work Experience

Backend Software Developer (Werkstudent) 2025 – Present

Fraunhofer IEM, Paderborn

Backend development across two projects with Paderborn-Lippstadt Airport, working in multiple cross-functional teams working with backend, sensor technology and hardware integration.

AG4NBA: Automated Luggage Handling (current, M.Sc. thesis)

- Developing an optimization algorithm for automated luggage packing at Paderborn-Lippstadt Airport that maximises bin utilisation, using ~20% fewer containers per flight to lower transport costs from the baggage handling system(BHS) to the aircraft and reducing manual labor by roughly 30 hours per week.
- Designing and implementing a full-scale system from the sensor data management through the luggage tracking and placement algorithms to the robotic-arm integration, with a microservice-based backend in Python supported by a PostgreSQL.

FastGate: Aircraft Localisation

- Automated boarding-bridge docking, aiming to reduce time of manual marshalling per aircraft and eliminating the need for 2-3 ground staff per docking.
- Built point-cloud alignment modules achieving localisation accuracy within 5-10 cm, enabling docking without human involvement.

Research & Teaching Assistant (Werkstudent) 2023 – 2024

Empirical Software Engineering, Paderborn University

- Led tutorials and student guidance, while contributing to LLM-security research (literature review and evaluation) and supporting conference submissions and publications.

Software Engineer (Full-Time) 2021 – 2022

Persistent Systems, Pune, India

- Built and operated production database backend services, running data migration and replication across SQL, MySQL, Oracle and PostgreSQL for the Quest SharePlex project.
- Collaborated in an Agile team using Jira and Confluence.

Skills

Programming: Python, Java, SQL, JavaScript, HTML

Backend & APIs: REST API design & development, microservices architecture, FastAPI, asynchronous processing, distributed systems (foundational)

AI & Computer Vision: Machine Learning, Computer Vision, OpenCV, Open3D, point-cloud processing, image processing

Databases: PostgreSQL, MySQL, Oracle, file-based databases; data integrity, migration & replication (Quest SharePlex), query optimisation

Testing & QA: API testing (Postman, Karate), test-driven development (TDD), code reviews

Tools & DevOps: Docker, uv, Git, Jenkins (CI/CD), Maven, Kafka, Jupyter Notebook, Jira, Confluence; Agile/Scrum

Systems: Linux, RHEL 7.x/8.x, Windows, macOS

Certifications

- Java Development Certification, QSpiders (6-month program, 2020).
- Multi-Dimensional Skill Training, Persistent Systems (Internal Training).

Education

M.Sc. Computer Science 2022 – Present (expected 2026)

Paderborn University, Germany

- Focus: Software Engineering, Machine Learning, Data-Driven Innovation, Explainable AI.

B.E. Computer Engineering 2016 – 2020

University of Mumbai, India

- Focus: Software Engineering, Digital Image Processing, Human-Machine Interaction.

Languages

English – C2 (fluent) | German – B1 (intermediate)

Leadership & Communication

Vice President, Indian Student Association (Paderborn University) 2024 – 2026

- Led teams and organised events for 200+ students; primary liaison with university administration and international student committees.